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Measuring Intra-Industry Trade - I

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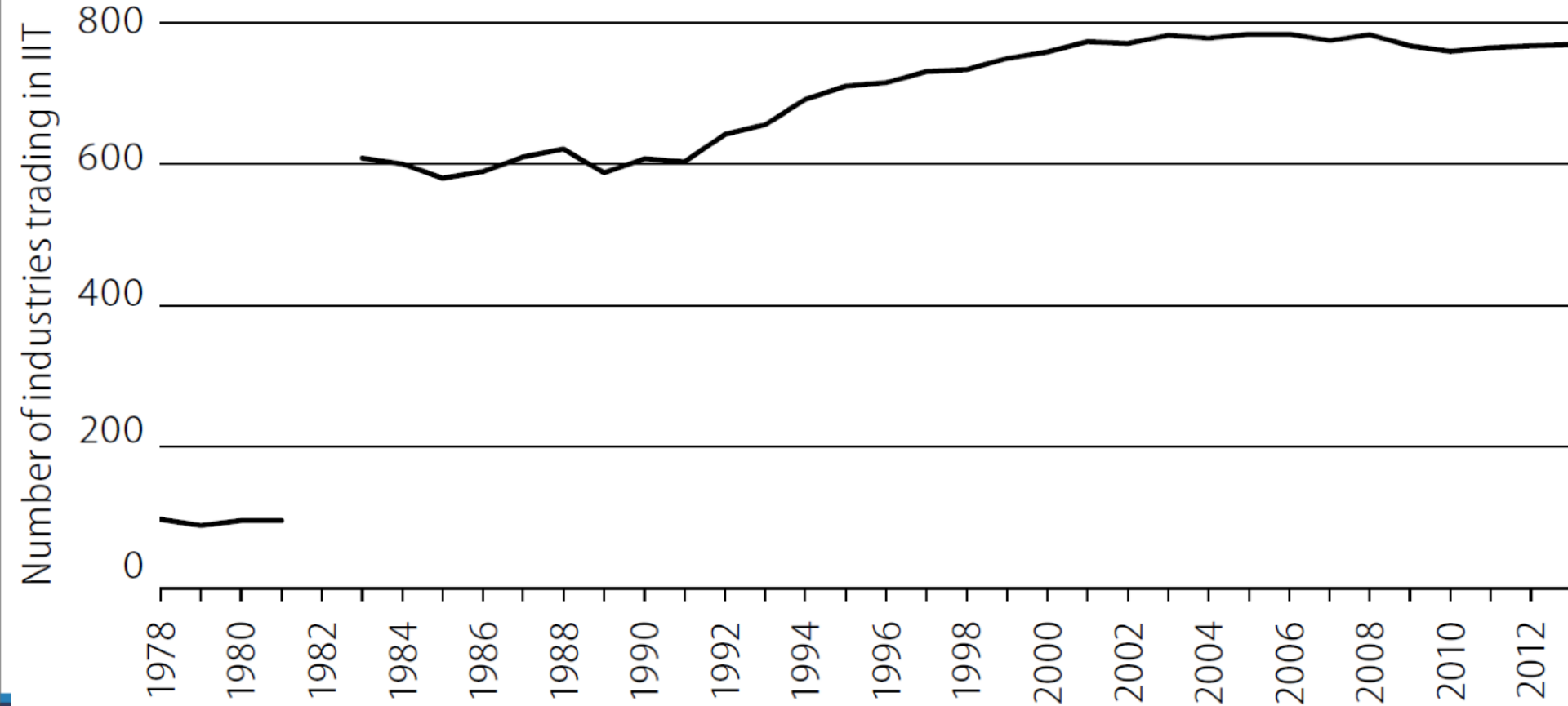
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Approaches to Measure

- Bagchi et al (2014) used the **unit value dispersion criterion** following Greenaway, Hine and Milner (**GHM**) approach at **HS-6-digit** level
- This GHM approach disentangles total IIT into horizontal and vertical,
- termed as, high vertical or ***h-VIIT*** and inferior, as, low vertical or ***l-VIIT*** products
- Bagchi et al (2014) estimated IIT for six major Indian manufacturing industries over the period 1990–2013.
- They found that technologically **inferior quality products** (or l-VIIT) have been dominant in India's export basket throughout the period of study,
- whereas HIIT and export of high technology goods (h-VIIT) have gained some momentum after the global economic **slowdown in 2008**

IIT in India



Data: World Bank's World Integrated Trade Solution (WITS)

GL measure

- Most commonly used measure of IIT was proposed by **Grubel and Lloyd (1975)**
- The Grubel and Lloyd (GL) index for any particular product class i between countries A and B is defined as follows:

where, X_{AB}^i is the value of exports of product class i by country A to country B , and M_{AB}^i is the value of imports of product class i by country A from country B

$$GL_{AB}^i = 1 - \frac{|X_{AB}^i - M_{AB}^i|}{X_{AB}^i + M_{AB}^i}$$

- The value of this index lies between **zero and one**, including the two extreme values.

GL Measure

- Intra-industry Trade Index (T):

$$T = 1 - \frac{|X - M|}{X + M}$$

- X = exports
 - M = imports
 - Numerator is absolute value
 - T ranges from 0 to 1
 - $T=0$ when nation only imports or exports the good
 - $T=1$ when exports = imports.
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- Balassa observed that trade volume of differentiated products increased within broad industrial classification. **Balassa used the term IIT in 1966.**
 - Grubel and Lloyd calculated the **T index for 10 industrial countries** in 1967 and found the range 0.30 to 0.66

GL Limitations

- Although widely used, the GL index is criticized on two grounds.
 - First, it does not take into account a country's trade imbalance.
 - second, it cannot distinguish between the type of IIT, that is, horizontal or vertical.
- To overcome the first limitation, Bergstrand (1983) proposed an index which adjusts for each country's multilateral trade imbalance in the following way:

$$BG_{AB} = 1 - \frac{|\widetilde{X}_{AB}^i - X_{BA}^i|}{\widetilde{X}_{AB}^i + X_{BA}^i}$$

$$\text{where, } \widetilde{X}_{AB}^i = \frac{1}{2} \left[\frac{X_A + M_A}{2X_A} + \frac{X_B + M_B}{2M_B} \right] X_{AB}^i \text{ and}$$

$$\widetilde{X}_{BA}^i = \frac{1}{2} \left[\frac{X_A + M_A}{2M_A} + \frac{X_B + M_B}{2X_B} \right] X_{BA}^i$$

- Grubel-Lloyd Uncorrected (GLU) formula, used for country j for industry i as the following:
- However, when the GLU index is applied for measuring the IIT between developed and developing countries, possibility of underestimation cannot be ruled out due to trade imbalance.
- The Grubel-Lloyd Corrected (GLC) formula involving country j for industry i , uses the following formulation:

$$GLU = \frac{\sum_i (X_{ij} + M_{ij}) - \sum_i |X_{ij} - M_{ij}|}{\sum_i (X_{ij} + M_{ij})} \times 100$$

$$GLC = \frac{\sum_i (X_{ij} + M_{ij}) - \sum_i |X_{ij} - M_{ij}|}{\sum_i (X_{ij} + M_{ij}) - \left| \sum_i X_{ij} - \sum_i M_{ij} \right|} \times 100$$

HIIT and VIIT

- **HIIT** → similar in terms of quality but have different characteristics or attributes, explained by the framework developed by Dixit and Stiglitz (1977), Lancaster (1980) and Krugman (1980, 1981).
 - Helpman and Krugman (1985) concluded that the larger is the difference in factor endowments, the smaller (larger) the extent of HIIT (VIIT).
- **VIIT** → trade in similar products of different qualities, which are not the same in terms of unit production costs and factor intensities (Falvey, 1981; Falvey and Kierzkowski, 1987)



- Vertical Intra-industry Trade (VIIT) dominates Horizontal Intra-Industry Trade (HIIT) for the selected countries like India.
- Bilateral IIT level with respect to India's high-income partners, while the same effect is non-significant for low-income countries

IIT and Rest of the World

Figure 1: India's Overall IIT with ROW (2001-2015)

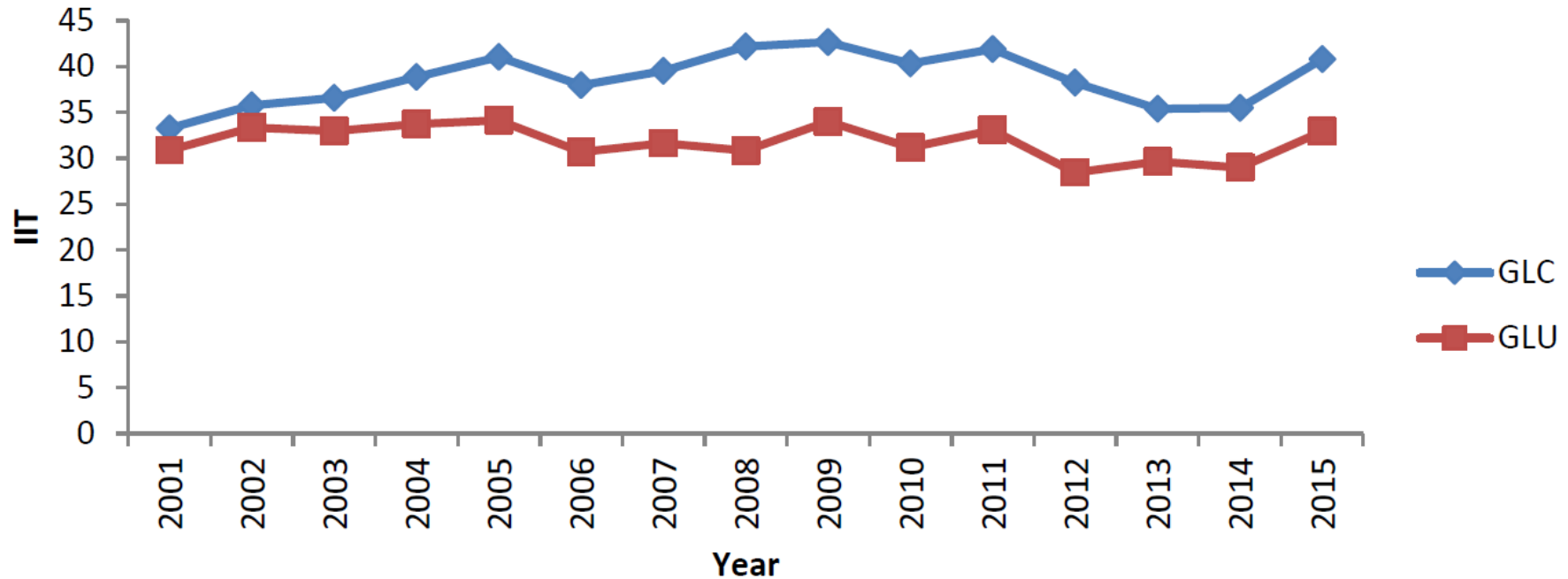


Table 2: Average Shares of India's Major Trade Partners in the Trade Basket

No.	Country	Export Share (%)			Import Share (%)		
		2001-05	2006-10	2011-15	2001-05	2006-10	2011-15
1	Australia	0.90	0.75	0.87	2.93	3.63	2.49
2	USA	18.50	12.51	12.98	6.38	6.41	4.87
3	China	4.41	6.46	4.67	5.30	10.66	12.52
4	Indonesia	1.47	1.61	1.67	2.26	2.38	3.20
5	Japan	2.97	2.11	1.95	3.22	2.53	2.37
6	Korea	1.24	1.89	1.43	2.86	2.76	2.88
7	Iran	1.14	1.22	1.18	0.43	3.80	2.28
8	South Africa	1.01	1.46	1.62	2.51	1.70	1.63
9	UK	4.78	3.80	3.06	4.11	1.88	1.35
10	Qatar	0.20	0.29	0.30	0.34	1.29	3.00
11	Malaysia	1.43	1.53	1.50	2.23	2.24	2.19
12	Thailand	1.25	1.06	1.12	0.77	0.98	1.22
13	Sri Lanka	1.77	1.51	1.66	0.24	0.19	0.15
14	Germany	3.82	3.14	2.54	3.83	3.81	2.97
15	Switzerland	0.73	0.36	0.40	4.91	4.71	5.59
16	Netherland	2.06	3.05	2.64	0.77	0.66	0.55
17	Singapore	3.53	4.46	4.02	2.47	2.62	1.66
18	Hong Kong	4.95	3.96	4.24	1.60	1.69	1.73
19	Vietnam	0.62	0.95	1.65	0.06	0.15	0.53
20	Bangladesh	2.22	1.43	1.73	0.09	0.11	0.13
21	Brazil	3.02	2.48	2.03	0.57	0.66	1.00
22	Belgium	0.72	1.40	1.83	4.90	2.08	2.22
23	Italy	2.66	2.27	1.63	1.34	1.40	1.02
24	Nigeria	0.93	0.79	0.89	0.10	2.93	2.96
25	France	2.08	1.85	1.70	1.55	1.68	0.80
	Total	68.41	62.34	59.31	55.77	62.95	61.31



Table 3: India's IIT Results for Top Trade Partners

Country	Intra Industry Trade Index			Partnership / Negotiations with India through Trade Bloc	Status
	2001-05	2006-10	2011-15		
Developed Economies					
Australia	11.25	12.84	7.69	CECA, RCEP	Under Negotiations
Belgium	62.88	50.72	51.98	India-EU BTIA	Under Negotiations
France	19.42	22.82	33.19	India-EU BTIA	Under Negotiations
Germany	25.57	35.39	40.10	India-EU BTIA	Under Negotiations
Hong Kong, SAR	60.05	64.29	57.84		No FTA
Italy	27.85	24.12	30.87	India-EU BTIA	Under Negotiations
Japan	13.03	18.05	19.56	JICEPA, RCEP	CEPA
Netherlands	23.91	24.88	25.49	India-EU BTIA	Under Negotiations
Qatar	1.22	7.11	15.11	GCC	Framework Agreement signed
Singapore	21.19	48.41	39.44	ISCECA, IASEAN FTA, RCEP	FTA, CECA
South Korea	17.71	29.90	38.03	IKCEPA, RCEP	CEPA
Switzerland	36.86	43.54	36.10	India-EFTA Agreement	Under Negotiations
UK	18.22	25.85	27.53	India-EU BTIA	Under Negotiations
USA	31.21	26.63	29.82		No FTA
Developing Economies					
Bangladesh	12.66	16.85	22.74	SAFTA, BIMSTEC	FTA
Brazil	6.69	10.28	7.76	India Mercosur PTA, IBSA	PTA
China	15.07	15.12	20.36	APTA, RCEP	PTA
Indonesia	11.74	14.36	13.22	IICECA, IASEAN FTA, RCEP	FTA
Iran	9.00	9.35	3.43	GSTP	No FTA
Malaysia	19.03	22.63	24.01	IMCECA, IASEAN FTA, RCEP	CECA
Nigeria	7.56	0.53	0.47	GSTP	No FTA
South Africa	4.97	5.64	3.97	IBSA, SACU PTA	Under Negotiations
Sri Lanka	29.80	30.77	43.33	ISLFTA, BIMSTEC	FTA
Thailand	20.35	25.61	30.29	BIMSTEC, IASEAN FTA, RCEP	FTA
Vietnam	10.55	16.04	12.87	IASEAN FTA, RCEP	FTA



- The importance of intra-industry trade became apparent after 1958 when tariff and other obstructions were removed among members of EU.
- Caused sharp fall in the unit cost of production in EU.
- Helpman, Krugman, Lancaster & others since 1979:
 - While trade (H-O model) based on comparative advantage is likely to be larger when the difference in factor endowments among nation is greater, intra-industry trade is likely to be larger among industrial economies of similar size and factor proportions.
 - Pretrade-relative commodity price may no longer predict the pattern of trade.
 - In contrast to H-O model, which predicts that trade will lower the return of the nation's scarce factor, with intra-industry trade, it is possible for all factors to gain. (this explains the formation of EU)
 - Intra-Industry Trade offers sharp increase in international trade in parts and components of products, or out-sourcing.



- Lancaster 1980: Even in the case of intra-industry trade, comparative advantage is somewhere in the background. Inter-industry trade reflects natural comparative advantage while intra-industry trade reflects acquired comparative advantage.

*Thank
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